SAI SAHITHI KOTHAPALLI

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Professional Summary

Results-driven AI Engineer with 2+ years of experience building enterprise AI applications using RAG, agentic AI, and LLM orchestration. Experienced in developing AI-driven workflow automation, intelligent assistants, and production-ready AI systems that streamline enterprise operations and reduce manual effort. Proficient in Python, LangChain, LangGraph, FastAPI, and modern LLM frameworks.

Work Experience

Software Engineer — AI/GenAI Jul 2024 – Present
Sify Technologies *Chennai, India*

- Worked on AI/GenAI projects for Bank of India (BOI Client), developing and deploying production-grade GenAI applications focused on agentic workflows, RAG pipelines, and LLM orchestration.
- Collaborated with cross-functional teams to integrate AI models into scalable enterprise-level solutions, improving operational efficiency and compliance automation.
- Conducted research and prototyping to integrate and scale AI solutions into enterprise infrastructure and standalone applications.

GenAI Intern Feb 2024 – Jun 2024
Sify Technologies *Chennai, India*

- Gained hands-on experience with Generative AI technologies, tools, and frameworks, including model inference and deployment workflows.
- Developed practical proficiency with AWS cloud services and subsequently cleared the AWS Certified Developer – Associate certification through applied learning.

Projects

RAG-based Audit & Compliance Remediation Engine | *Agentic AI Application*

Python, FastAPI, Azure OpenAI, LangChain, LangGraph, FAISS, Groq, Docker

- Built a stateful agentic GenAI system for regulatory reasoning using RAG and multi-step workflows; implemented graph-based orchestration with conditional execution, retries, and tool calling.
- Developed high-throughput document processing workflows to synthesize disparate regulatory sources into optimized vector stores and structured knowledge repositories.
- Designed a cross-provider LLM orchestration framework to mitigate API rate limits and provider outages, achieving **99.5%+ service availability** through strategic redundancy.
- Enhanced retrieval quality by implementing advanced chunking and hybrid search strategies, delivering a **30% improvement** in system performance.
- Engineered an enterprise-grade evaluation and validation layer using confidence thresholds and system-prompt optimization to minimize hallucination risks.

AI Observability & Guardrails Engine for RAG Systems | *LLMOps & Evaluation*

Python, FastAPI, LangChain, LangGraph, Azure OpenAI, Groq, FAISS, Redis

- Built an observability and evaluation engine for RAG systems, monitoring LLM responses, retrieval quality, and overall system performance in real time.
- Reduced hallucinations by **25%+** using grounding checks, confidence scoring, and LLM-based evaluation (LLM-as-a-judge).
- Implemented guardrails combining LLM-as-a-judge with rule-based validation, lowering invalid responses by **30%**.
- Enabled end-to-end pipeline tracing, improving debugging time by **40%**, and monitored multi-LLM usage across Azure OpenAI and Groq to optimize latency, reliability, and cost.

Ticket Resolution Automation System | *Agentic AI Application*

LangGraph, Groq, ServiceNow, n8n, Python

- Architected an agentic AI workflow using LangGraph (Python) to automate ticket resolution within Sify's IT management ecosystem, introducing intelligent state-based decision-making.
- Orchestrated automated triage, root-cause analysis, and resolution routing using Groq-powered LLMs, significantly reducing mean time to resolution (MTTR) for recurring incidents.

- Built n8n automation pipelines to bridge the agentic AI layer with enterprise ServiceNow workflows, enabling seamless handoff between AI-driven decisions and human escalation paths.
- Implemented stateful conversation graphs with LangGraph to handle multi-turn diagnostic reasoning and conditional escalation logic across complex incident types.

Agentic AI Firewall Audit System | *Agentic AI Application*

MCP, LangGraph, Claude/OpenAI, FortiGate REST API, Check Point API, Tool Calling, Python

- Built a fully agentic firewall audit application using Model Context Protocol (MCP) and LangGraph, enabling AI-driven security policy analysis across FortiGate and Check Point firewalls.
- Implemented tool calling to dynamically invoke FortiGate REST API and Check Point Management API endpoints, extracting firewall rules, policy configurations, and compliance posture in real time.
- Designed stateful audit workflows using LangGraph graphs with conditional branching, enabling the agent to reason across multi-vendor environments and generate consolidated compliance reports.
- Leveraged MCP server architecture to expose firewall audit tools as structured, reusable AI-callable functions, improving modularity and enabling seamless LLM-driven orchestration.

Skills

Languages & Backend: Python, FastAPI, Async APIs, Pydantic, REST APIs

GenAI & Agentic Systems: LangChain, LangGraph, RAG, Prompt Engineering, Tool Calling, Guardrails, Agent Workflows, MCP

LLMs & Orchestration: Azure OpenAI, Groq, Claude, Multi-LLM Routing, Token Optimization, Prompt Versioning

Retrieval & Data: FAISS, Vector Databases, Embeddings, Hybrid Search, Re-ranking, Chunking Strategies

ML & Frameworks: TensorFlow, PyTorch, Hugging Face Transformers

Evaluation & LLMOps: LLM Evaluation (LLM-as-a-judge), Observability, Tracing, Confidence Scoring

Cloud: Azure (Azure OpenAI, AI Search, Blob Storage, Cognitive Services, Azure Machine Learning), AWS, GCP

Systems & MLOps: Docker, CI/CD, Monitoring, Logging, Deployment, Redis Caching

Education

Sagi Rama Krishnam Raju Engineering College — Bhimavaram, Andhra Pradesh
Bachelor of Technology in Computer Science Engineering; CGPA: 8.75/10 — June 2020 – April 2024

Certifications

- Google Cloud Certified: Generative AI Leader
- Oracle Cloud Infrastructure Certified Data Science Professional
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- AWS Certified Developer – Associate